FMoA-RG: Frequency-guided Mixture of Adapters for Multimodal Image Fusion Report Generation

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***Abstract*—Cross-modal image-to-text report generation is an important research direction at the intersection of computer vision and natural language processing, aiming to automatically produce detailed and human-like textual reports from multimodal visual data. However, existing methods often suffer from information fragmentation, where descriptions from different modalities remain disjointed, making it difficult to achieve holistic semantic integration and resulting in incomplete and incoherent reports. To address this issue, we propose the Frequency-guided Mixture of Adapters for Report Generation (FMoA-RG) framework, which enables deep and adaptive fusion of multimodal images to generate semantically consistent and information-rich reports. Built upon a pretrained Vision Transformer (ViT), the proposed method leverages a frequency-guided dynamic expert routing mechanism to enhance fine-grained detail perception and adopts an adaptive dual-branch fusion strategy to optimize cross-modal feature integration and report generation. Experimental results demonstrate that, in typical multimodal fusion tasks such as multi-focus, multi-exposure, and infrared–visible image fusion, our method effectively mitigates information fragmentation in multimodal report generation, significantly enhances the completeness of details and semantic coherence of the generated reports, and exhibits strong generalization ability and practical applicability.**

I. Introduction

Cross-modal image-to-text report generation, a specialized domain at the intersection of computer vision and natural language processing, has seen significant advances, particularly in high-stakes fields like medical diagnostics and industrial quality control[1]. The objective of this field is to automate the creation of detailed, human-like textual reports from complex visual data, thereby improving efficiency and diagnostic accuracy. The evolution of this technology has been marked by a clear architectural progression. Early models were predominantly based on encoder-decoder frameworks, which later incorporated attention mechanisms to focus on salient image regions during text generation.More recently, Transformer-based architectures have become the state-of-the-art, leveraging self-attention to model long-range dependencies within each of visual and textual data, leading to more coherent and contextually accurate reports[2, 3].

To achieve a more holistic understanding of a scene, especially in complex diagnostic scenarios, recent research has increasingly focused on leveraging multimodal data, fusing information from different imaging sources to create a richer input representation. For instance, in medical imaging, studies have combined structural data from MRI with functional data from PET scans to improve diagnostic precision[4]. In remote sensing and autonomous driving, the fusion of RGB and thermal infrared images has been shown to improve object detection under varying environmental conditions[5]. A similar strategy has also proven valuable in industrial defect analysis, where integrating 2D surface images with 3D point cloud data enables not only the detection of anomalies but also the characterization of their depth and shape, providing critical information for root cause analysis and process optimization[6]. These approaches aim to harness the complementary strengths of each modality, compensating for the limitations of any single data source.

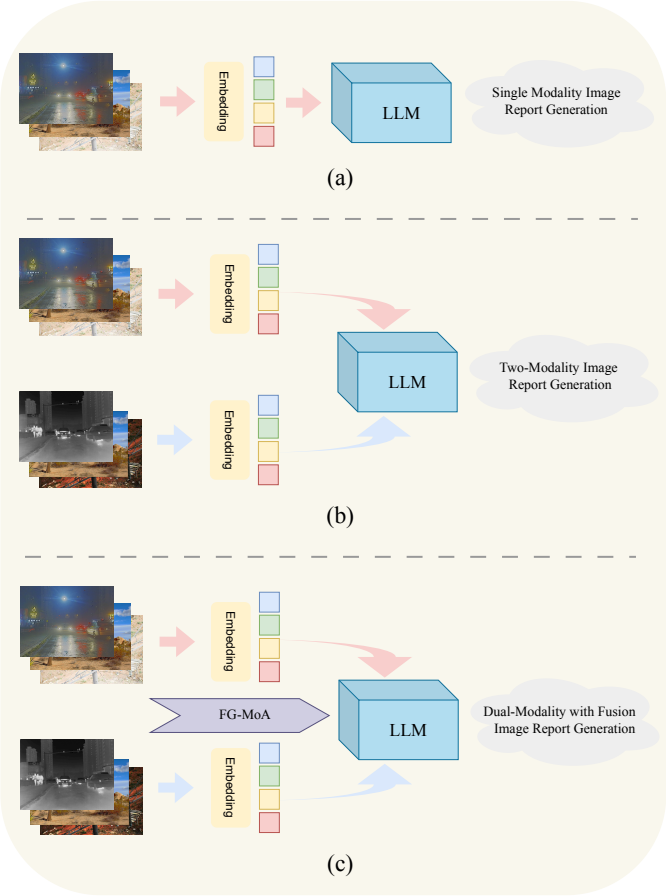


Fig. 1. Report generation under different conditions: (a) report generation from single-modal images, (b) report generation from dual-modal images, and (c) report generation under dual-modal image fusion, where the generated reports provide more accurate descriptive information.

Despite these efforts, a fundamental challenge persists in the fusion process itself. Existing cross-modal image-text report generation methods often prioritize the expression of information from individual modalities, generating descriptions for each feature set separately rather than achieving a deep,

synergistic fusion of the multimodal data. This leads to a phenomenon known as "information fragmentation," where the resulting textual reports are disjointed and fail to provide a comprehensive, systemic summary of the rich information contained within the cross-modal images. The generated text, while potentially accurate for each modality in isolation, lacks the integrated insight that a human expert would derive from a holistic analysis, thus limiting its practical utility.

To address these shortcomings, we introduce a novel framework: the Frequency-guided Mixture of Adapters for Report Generation (FMoA-RG). Our framework is designed to facilitate the deep fusion of multimodal images to generate reports that are both detailed and holistically coherent. Built upon a pre-trained Vision Transformer (ViT) as its encoder-decoder backbone for robust feature representation, our approach introduces three key innovations:

- We have proposed a universal multimodal fusion-enhanced text report generation framework, which can adapt to various complex scenarios and information deficiencies, and effectively encodes and decodes multimodal information.
- We design a novel Frequency-Guided Mixture of Adapters (FG-MoA) module, which utilizes a gated Mixture-of-Experts (MoE) routing mechanism guided by high-frequency information to dynamically adjust attention, thereby enhancing the model's perception of fine-grained details.
- We introduce a dual-branch fusion optimization strategy with automatic selection to integrate results from various fusion routes, automatically choosing the best scheme to maximize the accuracy and completeness of the final report.

In summary, our FMoA-RG framework directly addresses the critical challenge of information fragmentation in multimodal report generation. By enabling a deep and adaptive fusion of cross-modal data, it effectively leverages complementary features to enhance the semantic richness and consistency of the output. The result is the generation of detailed, coherent reports that contain synthesized, actionable insights, making them highly valuable for diverse and complex application scenarios.

II. Related Work

A. Image Report Generation Technology

Current deep learning-based image report generation predominantly employs end-to-end vision-language model (VLM) architectures. Mainstream approaches such as BLIP-2 [7] and Flamingo [8] achieve direct mapping from images to text through large-scale vision-language pre-training. Although these models demonstrate proficient performance in single-modal image description tasks, their generated reports often suffer from incomplete information and misplaced emphasis in complex scenarios requiring integration of multi-source data. These shortcomings primarily stem from the absence of effective multimodal information fusion mechanisms capable of comprehensively consolidating disparate data sources.

To address this limitation, researchers have introduced a "fusion-then-generation" paradigm [9]. This methodology first fuses multiple source images using multimodal fusion techniques, integrating complementary information scattered across different modalities into a single, information-enhanced representation characterized by higher density and integrity. These enhanced representations serve as input for subsequent language models, substantially improving report generation quality. Yao et al. [10] validated the effectiveness of this approach in medical imaging through multimodal instruction tuning. The latest Reg2RG framework [11] further incorporates a region-guided mechanism, enhancing the model's capacity to perceive and describe clinically relevant regions, thereby strengthening the technical foundation for generating reports with greater clinical value.

B. Image Fusion and Frequency-Guided Adapter Mechanism

General image fusion techniques seek to develop versatile unified frameworks capable of handling multi-modal, multi-exposure, and multi-focus scenarios within a single architecture. Representative works U2Fusion [12] and SwinFusion [13] adopt shared backbone networks and unified optimization objectives, achieving multi-functional fusion by learning cross-task common features. Nevertheless, while pursuing generality, this approach fails to address specific task requirements, frequently leading to loss of fine-grained details.

To achieve finer-grained fusion, Zhu et al. [14] introduced the idea of the mixture of experts (MoE) and proposed the Task-Customized Mixture of Adapters (TC-MoA) model, which dynamically activates different adapters for distinct fusion tasks. However, TC-MoA's routing mechanism relies solely on task labels, representing macro-level control that is insensitive to local variations in image content.

Our core contribution is the insight that truly effective information enhancement must be driven by the image content itself. To this end, we propose Frequency-guided Mixture of Adapters for Multimodal Image Fusion Report Generation (FMoA-RG). Given that high-frequency components precisely correspond to critical image details such as edges and textures [15, 16], our method leverages the high-frequency energy distribution as a routing signal to dynamically select and combine the most suitable adapters at the pixel level. This enables the fusion process to adaptively focus on and enhance local structures most valuable for downstream report generation tasks. Consequently, our approach fundamentally surpasses prior fusion methods that rely on task labels or generic feature learning, achieving finer-grained and more semantic-aware information enhancement.

III. METHODS

A. Overview

In this work, we propose a novel model termed a frequency-guided mixture of adapters for multimodal image fusion report generation (FMoA-RG), specifically designed for the task of image report generation. As illustrated in Fig. 2, the overall framework is composed of three key components: a Frequency-Guided Mixture of Adapters (FG-MoA), a Spatial

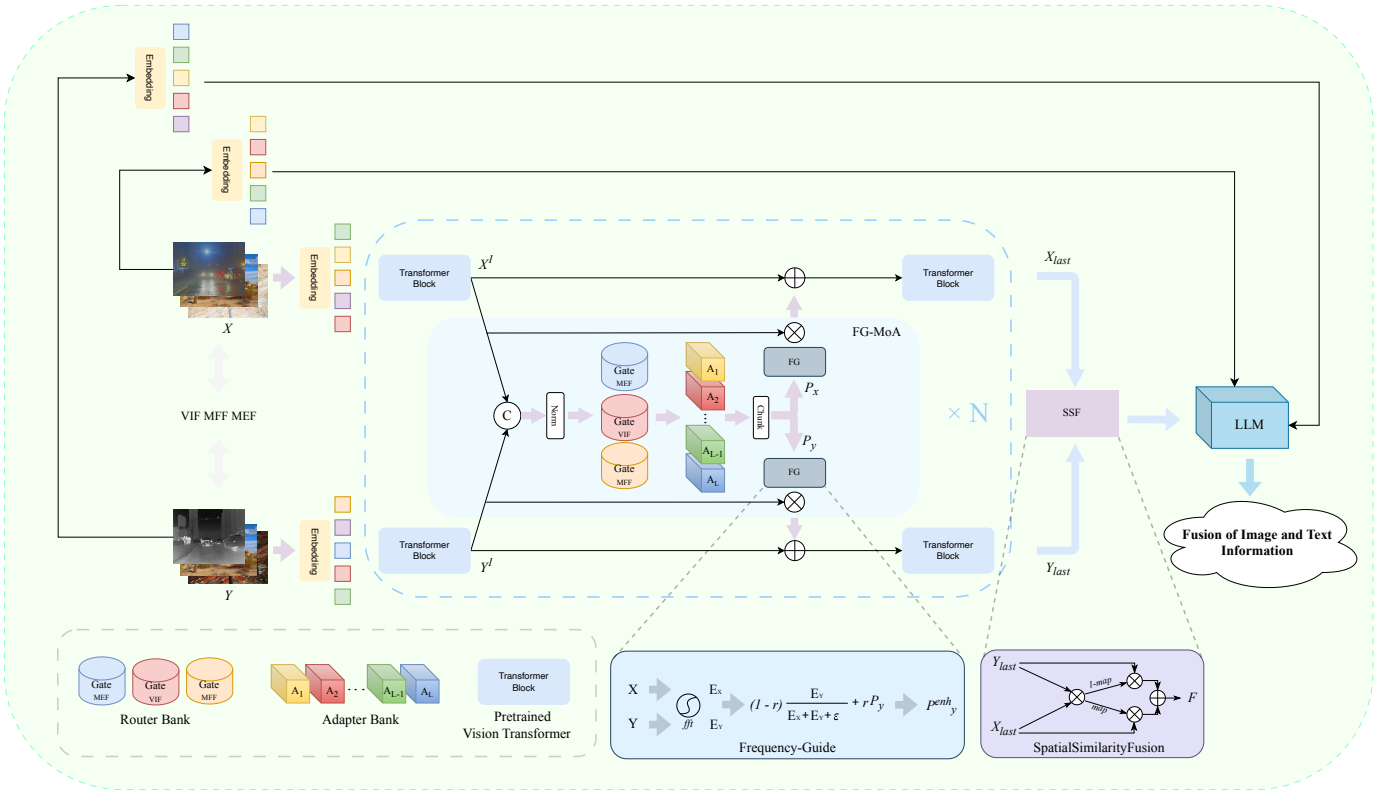


Fig. 2. Overview of the Frequency-guided Mixture of Adapters for Multimodal Image Fusion Report Generation(FMoA-RG), including the requency-Guided Mixture of Adapters and SpatialSimilarityFusion Module structure.

Similarity Fusion network, and an LLM-based image description generation structure.

The pipeline begins by embedding the two input modality images into feature vectors. These embeddings are subsequently processed through a pretrained Vision Transformer block, yielding their respective feature representations. At this point, the two representations are fed into the FG-MoA module. Within FG-MoA, the representations first pass through a Router Bank, which, guided by the fusion scenario, dynamically selects appropriate routers to obtain the corresponding adapter weights. With these weights, an L-way Mixture of Adapters (MoA) mechanism performs dynamic attention adjustment, thereby generating intermediate fusion features. Building on this, the Frequency-Guide module refines the fusion features by adaptively incorporating high-frequency information, ensuring that critical fine-grained details are effectively preserved. By stacking N layers of FG-MoA, the model progressively enhances the quality of the fused features.

After exiting FG-MoA, the Spatial Similarity Fusion module computes similarity-aware fusion weights between the two modalities and performs adaptive weighted fusion, ultimately producing the final fused output.

Finally, the pretrained LLM takes the fused output alongside the two original modality inputs, enabling cross-modal complementation and mutual enhancement. This joint input design significantly enriches both the semantic expressiveness and consistency of the generated reports, leading to more comprehensive and coherent descriptions.

B. Frequency-Guided Mixture of Adapters

By performing embedding extraction to obtain multimodal embeddings from the input images $X, Y \in \mathbb{R}^{H\times W\times C}$, we obtain the visual embeddings $f_X = \mathcal{V}(X) \in \mathbb{R}^{N_v\times d_v}$ and $f_Y = \mathcal{V}(Y) \in \mathbb{R}^{N_v\times d_v}$, where N_v denotes the sequence length and d_v represents the embedding dimensionality.

Afterwards, the two visual embeddings are concatenated to obtain a fused visual embedding, which facilitates the gating selection for the adapters:

$$f = \text{Concat}(\{f_X\}, \{f_Y\}) \in \mathbb{R}^{N_v\times 2d_v} \tag{1}$$

Following normalization and dimensionality reduction, we select the corresponding gating function G_S from the Router Bank according to the specific scenario S. The gating network G_S generates a sparse selection vector $g_s(f) \in \mathbb{R}^{N_p}$, which assigns the primary soft weights to the N_p adapters. This sparse selection is computed by applying a linear transformation to the input features, injecting Gaussian noise, and subsequently performing a Softmax followed by a TopK operation:

$$g_s(f) = \text{TopK}\left(\text{Softmax}\left(W_s Norm(f) + b_s + \mathcal{N}(0,\sigma^2)\right), K\right) \tag{2}$$

In this process, the role of TopK is to select the k largest values in $F(x)$ and set all other values to zero. Formally, it can be defined as:

$$\text{TopK}(F(x),k)_i = \begin{cases} F(x)_i, & \text{if } F(x)_i \in \text{Top-k}(F(x)) \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

After this step, for the initialized gating weights $g_s(f)$, we perform normalization over the non-zero entries to obtain the final gating weights $G_s(f)$. Specifically, the weight for the i-th adapter is computed as follows:

$$G_s(f)_i = \begin{cases} \dfrac{\exp(g(f)_i)}{\sum_{j\in\mathcal{I}} \exp(g(f)_j)}, & \text{if } i \in \mathcal{I} \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

where $\mathcal{I} = \{i \mid g(f)_i^t \neq 0\}$ denotes the set of selected adapter indices (i.e., those retained after applying TopK).

After obtaining the weights of each Adapter, we perform a weighted summation over the outputs of all Adapters to derive the final fused result, which can be formulated as:

$$F_{\text{out}} = \sum_{i=1}^{N_p} G_s(f)_i \cdot A_i(f), \quad F_{\text{out}} \in \mathbb{R}^{N_v \times 2d_v} \tag{5}$$

where $A_i(f)$ denotes the transformation result of the i-th Adapter given the input feature f, and $2d_v$ represents the output embedding dimensionality of the Adapters.

Afterwards, F_{out} is split into P_x and P_y to participate in the Frequency-Guide transformation that emphasizes high-frequency information:

$$F_{\text{out}} = \left[\, P_x,\, P_y \,\right], \quad P_x, P_y \in \mathbb{R}^{N_v \times d_v} \tag{6}$$

The Frequency-Guide module adjusts the fused embeddings $P_x, P_y \in \mathbb{R}^{N_v \times d_v}$ according to the high-frequency content of the original input images $X, Y \in \mathbb{R}^{H\times W\times C}$. First, the high-frequency energy of each input image is computed using a Fourier transform:

$$E_X = \sum_{c=1}^{C}\sum_{h=1}^{H}\sum_{w=1}^{W} \left|\text{FFT}(X)_{c,h,w}\right| \cdot \text{HF}_{h,w}, \tag{7}$$

$$E_Y = \sum_{c=1}^{C}\sum_{h=1}^{H}\sum_{w=1}^{W} \left|\text{FFT}(Y)_{c,h,w}\right| \cdot \text{HF}_{h,w}. \tag{8}$$

Here, $\text{HF}_{h,w}$ is a high-frequency mask retaining the outer frequency components, and $E_X, E_Y \in \mathbb{R}$ are scalar values.

The normalized frequency-guided weights are computed as:

$$w_X = \frac{E_X}{E_X + E_Y + \varepsilon}, \quad w_Y = \frac{E_Y}{E_X + E_Y + \varepsilon}. \tag{9}$$

Next, the embeddings are modulated via linear interpolation controlled by a hyperparameter γ:

$$P'_x = \gamma \cdot P_x + (1-\gamma)\cdot w_X, \quad P'_y = \gamma \cdot P_y + (1-\gamma)\cdot w_Y. \tag{10}$$

Here, $P'_x, P'_y \in \mathbb{R}^{N_v \times d_v}$, and the scalars $w_X, w_Y \in \mathbb{R}$ are broadcasted across the embedding dimensions.

These prompts are then applied to the embeddings via element-wise multiplication, optionally followed by a learned shift $\Delta_x, \Delta_y \in \mathbb{R}^{N_v \times d_v}$:

$$O_X = P'_x \odot X + \Delta_x, \quad O_Y = P'_y \odot Y + \Delta_y. \tag{11}$$

Finally, $O_X, O_Y \in \mathbb{R}^{N_v \times d_v}$ are fed into the subsequent Transformer block, where the same operations are iteratively repeated across n stacked layers:

$$\left(O_X^{(l+1)}, O_Y^{(l+1)}\right) = \mathcal{T}^{(l)}\left(O_X^{(l)}, O_Y^{(l)}\right), \quad l = 1, 2, \ldots, n. \tag{12}$$

C. SpatialSimilarityFusion

After the FG-MoA module, we obtain two streams of token-level features X_{last} and Y_{last}, where $X_{last}, Y_{last} \in \mathbb{R}^{N_v \times d_v}$, with N_v tokens and embedding dimension d_v. These token embeddings already encode semantic fusion and frequency-guided enhancement, yet they lack explicit spatial structure. Mapping them back to the image domain is necessary to enable pixel-level refinement and preserve local high-frequency details that may have been diluted during token aggregation.

A decoder then linearly projects each token into a patch-level pixel vector:

$$Z_X = W_d X_{last}, + b_d, \qquad Z_Y = W_d\, Y_{last} + b_d, \tag{13}$$

where $Z_X, Z_Y \in \mathbb{R}^{N_v \times (p^2 C)}$, with each patch consisting of $p \times p$ pixels and C channels. This step allows the network to learn an optimal mapping from semantic token features to local pixel structures, making the representations suitable for image reconstruction while retaining the information captured by FG-MoA.

Next, the patch vectors are rearranged into a 2D spatial grid via the unpatchify operation:

$$x = \text{Unpatchify}(Z_X), \qquad t = \text{Unpatchify}(Z_Y), \tag{14}$$

where $x, t \in \mathbb{R}^{B\times C\times H\times W}$. This step restores the spatial layout, which is critical because pixel-level fusion requires explicit spatial correspondence.

If the two decoded branches have different spatial resolutions, the auxiliary branch is resized to match the primary branch:

$$t_q = \text{Resize}(t; H, W), \tag{15}$$

ensuring $t_q \in \mathbb{R}^{B\times C\times H\times W}$. This alignment guarantees a one-to-one correspondence between pixels, which is essential for computing meaningful per-pixel similarity.

A pixel-wise similarity map is then computed to measure local agreement between the two branches:

$$s = \sum_c x_c\, t_{q,c}, \quad \text{sim_map} = \sigma(s), \tag{16}$$

with $\text{sim_map} \in (0,1)^{B\times 1\times H\times W}$. The inner product across channels captures the alignment of semantic content at each pixel, while the sigmoid nonlinearity converts this measure into a smooth, interpretable weight.

Finally, Spatial Similarity Fusion (SSF) performs a weighted interpolation based on the similarity map:

$$F_{\text{fused}} = \text{sim_map} \odot x + (1 - \text{sim_map}) \odot t_q \tag{17}$$

yielding $F_{\text{fused}} \in \mathbb{R}^{B\times C\times H\times W}$. This design allows the fused output to selectively preserve high-confidence regions from

TABLE I
PERFORMANCE (%) OF DIFFERENT MODELS ACROSS THREE SCENARIOS. THE BEST RESULTS ARE IN **BOLD**, AND THE SECOND-BEST ARE UNDERLINED.

Score (%)	VIF					MFF				MEF		
	SwinFusion	LFDT-Fusion	PSLPT	CDDFuse	**Ours**	SwinFusion	LFDT-Fusion	PSLPT	**Ours**	LFDT-Fusion	PSLPT	**Ours**
BLEU-1	66.6	53.7	61.1	57.4	**73.6**	43.2	35.6	39.2	**54.6**	41.0	43.9	**50.9**
BLEU-2	46.2	33.4	37.2	36.1	**57.1**	33.3	28.4	21.6	**39.5**	**26.5**	21.8	28.5
BLEU-3	35.6	22.0	23.7	27.1	**48.2**	26.6	22.4	13.1	**29.1**	12.1	**19.3**	14.9
BLEU-4	28.1	14.3	16.7	21.0	**36.1**	**20.8**	16.8	6.7	20.6	2.3	3.8	**5.2**
ROUGE-1	61.9	51.3	**70.5**	54.5	65.8	45.9	43.4	54.5	**65.8**	45.5	51.4	**54.3**
ROUGE-2	28.6	18.4	19.5	18.7	**36.0**	36.1	32.1	26.7	**43.2**	10.1	13.6	**20.0**
ROUGE-L	45.2	38.5	45.6	36.4	**56.5**	45.9	41.0	41.6	**65.8**	25.7	30.5	**34.8**
METEOR	71.3	49.8	44.8	45.5	**76.4**	41.3	43.9	40.4	**65.9**	27.9	42.5	**47.9**

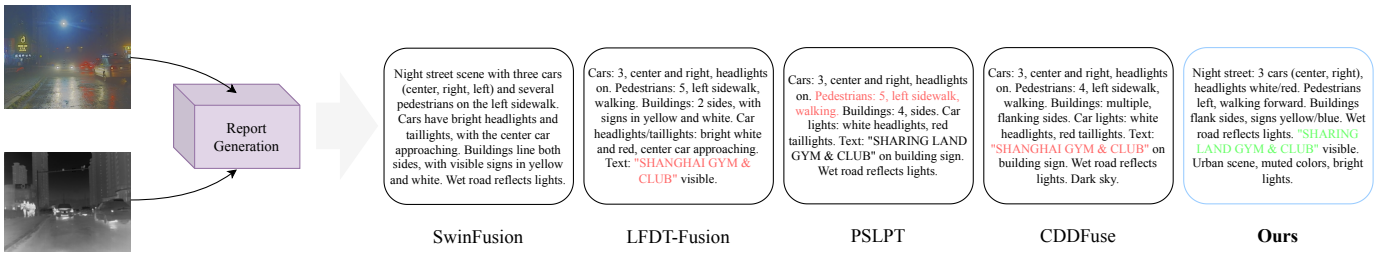


Fig. 3. The impact of different fusion modules on report generation under the VIF scenario, with red text indicating inconsistency and green text indicating consistency with the ground truth.

the primary branch while incorporating complementary information from the auxiliary branch, maintaining token-level semantic consistency and recovering fine-grained pixel-level details.

IV. EXPERIMENT

A. Dataset

To evaluate the proposed FMoA-RG method, we conducted experiments on three datasets: MSRS [17], a new multi-spectral dataset for infrared and visible image fusion, MFI-WHU [18], a multi-focus image fusion dataset, and the WireFO-MEF dataset we construct for Multi-Exposure Image Fusion.

MSRS [17] is built on the MFNet dataset [19]which contains 1,569 image pairs (820 taken at the daytime and 749 taken at nighttime). After removing misaligned image pairs and applying an image enhancement algorithm based on dark channel prior [20] to optimize the contrast and signal-to-noise ratio of infrared images, the final dataset contains 1,444 pairs of high-quality, aligned infrared and visible images with high quality.

MFI-WHU contains 120 image pairs from dataset provided by Cai et al. [21] and the public COCO dataset [22], aussian blur and a handmade decision map are added to generate multi-focus image pair.

WireFO-MEF is a large-scale dataset we collected in power grid construction scenarios. The images mainly focus on specific cases of foreign objects on power lines in such environments, which challenges the model's ability to capture fine-grained details. By adjusting the exposure levels of the original images, we generate synthetic overexposed and underexposed versions. Since these overexposed and underexposed images are derived from the same source image, they are perfectly aligned and thus require no image registration. The dataset contains images of various resolutions, most of which are of relatively high quality.

B. Evaluation Metrics

To evaluate the performance of our model on cross-modal image-to-text report generation, we adopt the Bilingual Evaluation Understudy (BLEU), Recall-Oriented Understudy for Gisting Evaluation (ROUGE), and Metric for Evaluation of Translation with Explicit Ordering (METEOR) as the primary evaluation metrics. These metrics collectively assess the lexical, phrasal, and semantic overlap between the candidate text and the reference text, thereby reflecting the accuracy, coverage, and fluency of the generated reports. Their formal definitions are summarized as follows.

First, the Bilingual Evaluation Understudy (BLEU) score focuses on the precision of n-grams in the candidate text with respect to the reference text. Specifically, the BLEU-N score for candidate text C and reference text R is given by:

$$\text{BLEU-N} = \text{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right) \tag{18}$$

where p_n denotes the modified n-gram precision of order n, w_n is the weight assigned to each n-gram order (typically uniform, i.e., $w_n = \frac{1}{N}$), and BP is the brevity penalty designed to penalize overly short candidates.

In contrast to BLEU's emphasis on precision, the Recall-Oriented Understudy for Gisting Evaluation (ROUGE) highlights recall by measuring the overlap of n-grams or sequences

between the candidate and the reference text. The ROUGE-n score is formally defined as:

$$\text{ROUGE-}n = \frac{\sum_{S\in\text{Ref}}\sum_{g\in S}\min\left(C_C(g), C_S(g)\right)}{\sum_{S\in\text{Ref}}\sum_{g\in S}C_S(g)} \tag{19}$$

where g denotes an n-gram in the reference text S, $C_C(g)$ is the occurrence count of g in the candidate text, and $C_S(g)$ is the occurrence count of g in the reference text.

Finally, the Metric for Evaluation of Translation with Explicit Ordering (METEOR) balances both precision and recall of unigram matches while additionally accounting for stemming, synonymy, and word order. It is formulated as:

$$\text{METEOR} = F_\alpha \cdot (1 - P_{\text{frag}} \cdot \gamma) \tag{20}$$

where $F_\alpha = \frac{PR}{\alpha P+(1-\alpha)R}$ is the harmonic mean of precision P and recall R with weight α, and P_{frag} is the fragmentation penalty based on alignment chunks, modulated by parameter γ.

C. Experimental results

We compared our model against four state-of-the-art general-purpose multimodal image fusion methods, including SwinFusion [13], LFDT-Fusion [23], PSLPT [24], and CDDFuse [25]. In the comparison experiments, we replaced the modality fusion module of our model with these methods while keeping the same large model-based approach for image description generation, in order to evaluate how different fusion strategies influence the accuracy and fluency of report generation. In some scenarios, certain models do not support multimodal image fusion under the given conditions and were therefore excluded from the comparison experiments. The results are summarized in Table II, which presents the BLEU, ROUGE, and METEOR scores of different models across the infrared–visible image fusion, multi-focus image fusion, and multi-exposure image fusion scenarios.

Specifically, on the MSRS dataset, our model achieves leading performance across all metrics, with the only exception being ROUGE-1, where it slightly falls short (65.8%) compared to PSLPT [24]'s 70.5%. This demonstrates our method's strong capability in capturing and understanding fine details in infrared–visible scenarios. By effectively leveraging the distinct information present in the two original modalities and the fused modality, our approach enhances the level of detail in image descriptions, thereby making them more comprehensive and fluent.

On the MFI-WHU dataset, our model also achieves state-of-the-art performance. In comparison with three other general models, it only lags behind SwinFusion [13] by 0.2% on the BLEU-4 metric. Meanwhile, on metrics such as BLEU-1 and ROUGE-1, which reflect whether the basic semantics are correct and complete, our model surpasses the second-best by more than 10%. This demonstrates the strong ability of our model to capture fine-grained image details, as well as the significant contribution of the dual-branch selective fusion strategy to improving report accuracy and completeness.

For the WireFO-MEF dataset, based on the specific scenario of foreign objects on power lines, we particularly emphasize the completeness of the reports during testing, in order to accurately and thoroughly reflect this special case. As shown in Table II, our model achieves the highest scores on BLEU-1 and ROUGE-1, which reflect the integrity and detail completeness of the information, with a significant margin over the second-best results. At the same time, it also performs well on other metrics that represent sentence fluency. This demonstrates that our model enriches detail expression by integrating information from images with different exposure levels, thereby achieving a more comprehensive and detailed understanding of the images.

TABLE II
PERFORMANCE (%) OF DIFFERENT SETTINGS ACROSS THREE SCENARIOS. THE BEST RESULTS ARE IN **BOLD**, AND THE SECOND-BEST ARE UNDERLINED.

Score	Scenario	Original	Dual-modal	Triple-modal
BLEU-4	VIF	18.1	<u>26.3</u>	**36.1**
	MFF	11.9	<u>12.3</u>	**20.6**
	MEF	3.2	<u>3.3</u>	**5.2**
ROUGE-L	VIF	43.1	<u>48.3</u>	**56.5**
	MFF	48.2	<u>56.5</u>	**65.8**
	MEF	<u>39.4</u>	35.3	**47.9**
METEOR	VIF	54.2	<u>62.1</u>	**76.4**
	MFF	<u>57.7</u>	53.1	**65.9**
	MEF	35.6	<u>39.2</u>	**47.9**

We also provide the specific report descriptions generated by our fusion method and other fusion models under the VIF scenario, as shown in Fig. 3. All compared models are applied in the VIF scenario. Among them, CDDFuse and LFDT-Fusion produce incorrect textual descriptions in the figure, while PSLPT misreports the number of pedestrians. Compared with the expressions generated by SwinFusion, our method yields more detailed report descriptions.

Beyond this, we also conducted comparative experiments on different LLM guidance strategies across three datasets with varying numbers of modalities, while keeping parameter settings and other details consistent. We evaluated the quality of the generated texts under different modality configurations using BLEU-4, ROUGE-L, and METEOR, as shown in Table. II. After incorporating two-modality information and fused-modality guidance into the LLM for report generation, the quality of the generated reports improved by more than 10% compared with using only the *Dual modal* and *Original image*. This demonstrates the effectiveness of our method in leveraging complementary modality information to enrich details and ensure report completeness.

V. CONCLUSION

This paper presents a Frequency-guided Mixture of Adapters (FMoA-RG) framework for multimodal image fusion report generation, addressing the long-standing issue of information fragmentation in cross-modal image-to-text tasks. By introducing a frequency-guided expert routing mechanism and a dual-branch adaptive fusion strategy, the proposed

method enables fine-grained feature perception and dynamic semantic enhancement, achieving deep and coherent multimodal integration. Extensive experiments on infrared–visible, multi-focus, and multi-exposure fusion tasks demonstrate that FMoA-RG significantly outperforms existing methods in BLEU, ROUGE, and METEOR metrics, producing reports with superior detail completeness and semantic coherence. Future work will explore instruction-based report generation with large multimodal models to enable more controllable and human-aligned report synthesis across diverse real-world scenarios.

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